

dirac16complex in the author's primordial gravitational field

Lagrangian, covariant field equations, non-triviality [1], energy-momentum tensor operator,
equations of state and the field equations for a4[x4] (Revision record)

dirac16complex project (with Claude Opus 5.5)

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Abstract

This document records, for the fermion field `dirac16complex` coupled to the author's primordial gravitational field, the Lagrangian, its coupling through the canonical spin connection, the exact covariant field equations (Euler-Lagrange) written out component by component, the non-triviality test [1] with its proof, the self-consistency of the field equations, the energy-momentum tensor (as a Grassmann-algebra expression and as a normal-ordered operator after canonical quantisation in $4 + 4$ dimensions, with the Krein structure that this quantisation forces), the kinetic and potential energy, the energy density, the pressures and the equations of state, and the Einstein-Lovelock field equations for `a4[x4]` with `dirac16complex` as the source. Every formula is an output of Revision code (a WolframScript verifier and, for most statements, an independent sympy verifier) and is listed with the names of the checks that verify it. The main exact results are: the spin-connection term of the field equation is $\gamma^\mu \Omega_\mu = 3H\gamma^{(8)}$ for every a_4 (the time-direction terms of the three inflating and the three deflating directions cancel, the hidden-direction terms add), so the Euler-Lagrange equations contain a nonzero gravitational term for every $H > 0$, and the metric is never flat for $H > 0$; the canonical anticommutator $\{\Psi, \Psi^\dagger\} = B \delta^7 / \sqrt{|g|}$ with B of signature (8,8) forces an indefinite (Krein) inner product; homogeneous on-shell configurations have $\rho = mS + U$ and $p = SU' - U$ in every direction; the field equations for a_4 reduce to a constraint, an evolution equation $a_4'' F(a_4') = \kappa(p_3 - p_t)$, a hidden-direction equation and the condition $p_3 + p_t = 2p_8$. Section 15 states what is not claimed.

1 Scope, sources and status of the record

The binding plan is `Revision/SPEC.md`; the author's task is quoted in `Revision/README.md`. This document belongs to the new record under `Revision/` and uses nothing from the earlier stages of the repository: every statement below is an output of Revision code and is cited by the name of the check that verifies it. Numbers and formulas are taken only from the following files (verdicts as recorded in them when this document was written):

part	files	checks
algebra (SPEC section 2)	<code>Revision/algebra</code> <code>/reports/wolfram</code> <code>-algebra.json</code> , <code>Revision</code> <code>/algebra/reports/python-</code> <code>algebra.json</code>	Wolfram: 45 of 45 checks pass; sympy: 35 of 35 checks pass
field theory (SPEC sections 3, 4, 6)	<code>Revision/theory/</code> <code>field-theory.json</code> , <code>Revision/theory/reports/</code> <code>wolfram-field-th</code> <code>eory.json</code> , <code>Revision</code> <code>/theory/reports/</code> <code>python-field-the</code> <code>ory.json</code>	Wolfram: 84 of 84 checks pass; sympy: 70 of 70 checks pass

part	files	checks
field equations for a4 (SPEC section 5)	Revision/field_equations_a4/a4-equations.json, Revision/field_equations_a4/reports/wolfram-a4-report.json, Revision/field_equations_a4/reports/python-a4-report.json	Wolfram: 47 of 47 checks pass; sympy: 61 of 61 checks pass
pairing (SPEC section 9)	Revision/pairing/pairing-theory.json, Revision/pairing/reports/wolfram-pairing.json, Revision/pairing/reports/python-pairing.json	Wolfram: 99 of 99 checks pass; sympy: 64 of 64 checks pass

Each part has two verifiers that share no code: a WolframScript verifier (exact symbolic arithmetic) and a sympy verifier (an independent re-derivation; the gamma matrices are rebuilt from the author's formulas on both sides and compared entry by entry). Where only one engine verifies a statement, only that engine's check is cited. A check is cited in typewriter type, for example `nontriviality_1_dirac16complex` (Wolfram) or `gamma_mu_Omega_mu_equals_3H_gamma_x8` (sympy); in the field-theory reports the Wolfram names end in `_G` for `dirac16complex` (explicit Grassmann algebra) and the sympy names begin with `grassmann_`. The proofs are exact identities in the stated generality (symbolic H , $a_4(x_4)$, m , λ , and a general field configuration on its jet space); exact numerical examples are labelled as examples.

Status of the comparison record of the field-theory branch. The sympy report contains a comparison with the Wolfram side (`comparison_with_wolfram`): 27 of 27 formula records and 63 of 63 check pairs agree, recorded against a Wolfram run of 82 checks. The Wolfram report has 84 checks; the two that the comparison record does not cover are `EMT_trace_G` and `EMT_trace_C` (the trace of the energy-momentum tensor), whose statement the sympy checks `grassmann_trace_on_shell` and `commuting_trace_on_shell` verify independently. The Wolfram formula file was also regenerated after that comparison: a re-run of the sympy checker made while this document was written (not committed) reproduced its checks and formulas unchanged and marked one formula record, `energy_exchange`, as not agreeing, because its parser expects that equation in Wolfram equation syntax while the regenerated file states it in words; the equation itself is a passing check on both sides (`energy_exchange_equation`). This document therefore cites each formula by the checks of each engine separately and does not use the comparison record as evidence. Regenerating the comparison is a task of the field-theory branch.

Conventions (SPEC sections 1 and 2): coordinates $x_1, \dots, x_8$ as the author names them; x_1, x_2, x_3 are ordinary 3-space, x_4 is the time, x_5, x_6, x_7 are the three extra times (time-like, deflating exponentially as a_4 increases), x_8 is the hidden space direction; $z = 6Hx_8 \in (0, \pi/2)$, $H > 0$; $a_4 = a_4(x_4)$, $a'_4 = da_4/dx_4$; $\partial_a = \partial/\partial x_a$. Frame indices are aligned with the coordinates and written in parentheses: $\gamma^{(a)}$ is the frame gamma matrix of direction x_a . Throughout, $i \in \{1, 2, 3\}$ labels the 3-space directions and $t \in \{5, 6, 7\}$ the extra times.

2 The primordial gravitational field

The author's metric (SPEC section 1), with signature (4,4) (space-like x_1, x_2, x_3, x_8 ; time-like $x_4, \dots, x_7$):

$$ds^2 = e^{2a_4} \sin^{1/3} z (dx_1^2 + dx_2^2 + dx_3^2) - dx_4^2 \\ - e^{-2a_4} \sin^{1/3} z (dx_5^2 + dx_6^2 + dx_7^2) + \cot^2 z dx_8^2, \quad z = 6Hx_8 \in (0, \pi/2).$$

The scale factor of 3-space is $e^{a_4} \sin^{1/6} z$ (inflating as a_4 increases), that of the extra times $e^{-a_4} \sin^{1/6} z$ (exponentially deflating). The diagonal vielbein $e^a{}_\mu = f_a \delta^a_\mu$ with frame metric $\eta_{ab} = \text{diag}(+1, +1, +1, -1, -1, -1, -1, +1)$ has

$$f_1 = f_2 = f_3 = e^{a_4} \sin^{1/6} z, \quad f_4 = 1, \quad f_5 = f_6 = f_7 = e^{-a_4} \sin^{1/6} z, \quad f_8 = \cot z$$

and reproduces the metric entry by entry (`metric_is_the_authors`, `vielbein_inverse`; `sympy metric_from_vielbein_equals_SPEC`). The volume element is

$$\sqrt{|g|} = f_1 f_2 \cdots f_8 = \sin z \cot z = \cos z,$$

independent of x_4 : the inflation e^{3a_4} of 3-space and the deflation e^{-3a_4} of the extra times compensate (`sqrt_det_g_is_cos_z`, `sqrt_det_g_equals_cos_z`). $H = 0$ is not a member of the family but a degenerate limit ($g_{11} \rightarrow 0$, $g_{88} \rightarrow \infty$ at fixed x_8 ; `degenerate_at_H_0`); every statement below assumes $H > 0$ and $0 < z < \pi/2$.

The Levi-Civita connection has 25 independent nonzero symbols (`christoffel_count`, `christoffel_symmetric_metric_compatible`; formula key `christoffel_nonzero`):

$$\begin{aligned} \Gamma^{x_i}{}_{x_i x_4} &= a'_4, & \Gamma^{x_i}{}_{x_i x_8} &= H \cot z, \\ \Gamma^{x_t}{}_{x_4 x_t} &= -a'_4, & \Gamma^{x_t}{}_{x_t x_8} &= H \cot z, \\ \Gamma^{x_4}{}_{x_i x_i} &= e^{2a_4} \sin^{1/3} z a'_4, & \Gamma^{x_4}{}_{x_t x_t} &= e^{-2a_4} \sin^{1/3} z a'_4, \\ \Gamma^{x_8}{}_{x_i x_i} &= -e^{2a_4} H \frac{\sin^{4/3} z}{\cos z}, & \Gamma^{x_8}{}_{x_t x_t} &= e^{-2a_4} H \frac{\sin^{4/3} z}{\cos z}, \\ \Gamma^{x_8}{}_{x_8 x_8} &= -\frac{6H}{\sin z \cos z}. \end{aligned}$$

The mixed Ricci tensor is diagonal (`ricci_mixed_components`, `ricci_scalar`, `curvature_nonzero_flat_only_formally`):

$$R^{x_i}{}_{x_i} = a''_4 - 6H^2, \quad R^{x_4}{}_{x_4} = 6(a'_4)^2, \quad R^{x_t}{}_{x_t} = -a''_4 - 6H^2, \quad R^{x_8}{}_{x_8} = -6H^2,$$

$$R = 6((a'_4)^2 - 7H^2).$$

Since $R^{x_8}{}_{x_8} = -6H^2 < 0$ for every $H > 0$ and every function a_4 (a constant a_4 included), the metric is never flat for $H > 0$ (`never_flat_for_H_positive`).

3 The field dirac16complex: Clifford algebra, $\text{Pin}(4,4)$ and $\text{Spin}(4,4)$

dirac16complex is a field Ψ with 16 complex anticommuting components $\Psi_A(x)$, $A = 1, \dots, 16$ (elements of a Grassmann algebra; after quantisation, operators with canonical anticommutators, section 11). It transforms under a 16-dimensional irreducible representation of $\text{Pin}(4,4)$, the double cover of $\text{O}(4,4)$; for the transformations of determinant 1 it transforms under the direct sum of two inequivalent irreducible 8-dimensional representations of $\text{Spin}(4,4)$, the double cover of $\text{SO}(4,4)$. These representation facts are proved exactly below for the author's gamma matrices.

The gamma matrices are the author's real 16×16 matrices T16, rebuilt by Revision code from the author's formulas (the tau matrices of the notebook) in WolframScript and, independently, in Python, with the notebook's frame order mapped to the author's coordinates (SPEC section 2): $\gamma^{(8)} = \text{T16}[0]$, $\gamma^{(1,2,3)} = \text{T16}[1..3]$, $\gamma^{(4)} = \text{T16}[4]$, $\gamma^{(5,6,7)} = \text{T16}[5..7]$ (T16_block_form, coordinate_map; the two constructions agree entry by entry, fixture_comparison_gammas_json). Exact properties:

- Clifford relation $\{\gamma^{(a)}, \gamma^{(b)}\} = 2\eta^{ab} I_{16}$ for all 64 pairs; the squares are +1 for x_1, x_2, x_3, x_8 and -1 for $x_4, \dots, x_7$ (Clifford_relation, clifford_relation, clifford_relation_sympy). Every $\gamma^{(a)}$ is a real signed permutation matrix, symmetric for the space-like and antisymmetric for the time-like directions, $(\gamma^{(a)})^T = \eta_{aa} \gamma^{(a)}$ (reality, signed_permutation_matrices, symmetry_pattern).
- $C = \gamma^{(8)} \gamma^{(1)} \gamma^{(2)} \gamma^{(3)}$ (the notebook's sigma16) is real symmetric with $C^2 = 1$, and every $C \gamma^{(a)}$ is real antisymmetric, equivalently $C \gamma^{(a)} C^{-1} = -(\gamma^{(a)})^T$ (C_real_symmetric, C_square_d_identity, C_gamma_real_antisymmetric; sympy C_conjugation). The Dirac adjoint is $\bar{\Psi} = \Psi^\dagger C$ and $S = \bar{\Psi} \Psi$.
- The chirality $\Gamma = \gamma^{(8)} \gamma^{(1)} \dots \gamma^{(7)} = \text{diag}(-I_8, I_8)$ anticommutes with every $\gamma^{(a)}$, $\Gamma^2 = 1$, $[\Gamma, C] = 0$ (Gamma_definition, Gamma_diag, Gamma_anticommutes_with_gammas, Gamma_commutates_with_C; sympy chirality_diag).
- $B = -iC \gamma^{(4)}$ is Hermitian with $B^2 = 1$ and eigenvalues +1 and -1, eight each: signature (8,8) (B_definition, B_Hermitian, B_squared_identity, B_signature_8_8; sympy B_hermitian_involution_signature). B commutes with $\gamma^{(1)}, \gamma^{(2)}, \gamma^{(3)}, \gamma^{(4)}, \gamma^{(8)}$ and anticommutes with $\gamma^{(5)}, \gamma^{(6)}, \gamma^{(7)}$ (B_gamma_relations).
- The 28 generators $S^{ab} = \frac{1}{4}[\gamma^{(a)}, \gamma^{(b)}]$ satisfy the relations of $\text{so}(4,4)$ (S_Lorentz_algebra, S_lorentz_algebra), rotate the gammas as a vector (S_gamma_commutator, S_vector_action), satisfy $(S^{ab})^T C + C S^{ab} = 0$ so that $\bar{\Psi} \Psi$ is $\text{Spin}(4,4)$ -invariant (S_preserves_C), and commute with Γ (S_commutates_with_Gamma). The Hermitian form $\Psi^\dagger B \Psi$ is invariant only under the $\text{Spin}(4,3)$ that fixes x_4 (S_preserves_B_only_off_x4).

Representation theorem (exact). The 256 ordered products of gammas are linearly independent and span the full matrix algebra $\text{Mat}(16)$ (Clifford_basis_spans_full_matrix_algebra, clifford_products_span_M16), and the commutant of the gammas is one-dimensional ($\text{Pin}44_{\text{irreducible_commutant_dim_1}}$, $\text{pin_commutant_dimension_1}$): the 16-dimensional representation is irreducible under $\text{Pin}(4,4)$. The commutant of the 28 generators S^{ab} is two-dimensional, spanned by

the chiral projectors $(1 \mp \Gamma)/2$ (`Spin44_commutant_dim_2_chiral_projectors`, `spin_commutant_dimension_2`); each chiral half (rows 1 to 8 with $\Gamma = -1$, rows 9 to 16 with $\Gamma = +1$) is irreducible (`chiral_halves_irreducible`, `spin_halves_irreducible`); the spaces of intertwiners between the halves are zero (`chiral_halves_inequivalent_intertwiners_0`, `spin_halves_inequivalent`); the even subalgebra is $\text{Mat}(8) \oplus \text{Mat}(8)$ (`even_subalgebra_dimension`, `even_products_span_M8_plus_M8`). Every $\gamma^{(a)}$, the lift of a reflection (an element of $\text{Pin}(4,4)$ of determinant -1 in $\text{O}(4,4)$), exchanges the two halves (`Spin_Pin_reflection_swaps_halves`, `reflections_exchange_halves`). Hence under $\text{Spin}(4,4)$ the 16 splits into $8_- \oplus 8_+$, two inequivalent irreducible 8-dimensional representations, which the reflections combine into the one irreducible representation of $\text{Pin}(4,4)$.

4 Coupling through the canonical spin connection

The coupling to gravity uses the vielbein and the canonical spin connection (SPEC section 3): $\omega_\mu{}^a{}_b$ is defined by the vielbein postulate $\partial_\mu e^a{}_\nu - \Gamma^\lambda{}_{\mu\nu} e^a{}_\lambda + \omega_\mu{}^a{}_b e^b{}_\nu = 0$, $\omega_{\mu ab} = \eta_{ac} \omega_\mu{}^c{}_b$ is antisymmetric in a, b , and

$$\begin{aligned} \Omega_\mu &= \frac{1}{2} \omega_{\mu ab} S^{ab}, & D_\mu \Psi &= \partial_\mu \Psi + \Omega_\mu \Psi, \\ \gamma^\mu &= e_a{}^\mu \gamma^{(a)} = \gamma^{(\mu)} / f_\mu, & D_\mu \bar{\Psi} &= \partial_\mu \bar{\Psi} - \bar{\Psi} \Omega_\mu. \end{aligned}$$

(This is the canonical connection with the frame metric; the notebook's contraction of the mixed $\omega_\mu{}^a{}_b$, which lacks a metric factor, is not used.) The postulate holds for all 512 components and $\omega_{\mu ab} = -\omega_{\mu ba}$ (`vielbein_postulate` in both field-theory reports, `omega_antisymmetric`, `spin_connection_antisymmetric`). Exactly 12 components with $a < b$ are nonzero (`omega_components`; formula key `omega_nonzero`):

$$\begin{aligned} \omega_{x_i(i)(4)} &= a'_4 e^{a_4} \sin^{1/6} z, & \omega_{x_i(i)(8)} &= H e^{a_4} \sin^{1/6} z, \\ \omega_{x_t(4)(t)} &= -a'_4 e^{-a_4} \sin^{1/6} z, & \omega_{x_t(t)(8)} &= -H e^{-a_4} \sin^{1/6} z, \end{aligned}$$

and $\omega_{x_4} = \omega_{x_8} = 0$. The spinor connection is (`Omega_components`; sympy `Omega_x4_and_Omega_x8_vanish`)

$$\begin{aligned} \Omega_{x_i} &= \frac{1}{2} e^{a_4} \sin^{1/6} z (a'_4 \gamma^{(i)} \gamma^{(4)} + H \gamma^{(i)} \gamma^{(8)}) & (i = 1, 2, 3), \\ \Omega_{x_t} &= -\frac{1}{2} e^{-a_4} \sin^{1/6} z (a'_4 \gamma^{(4)} \gamma^{(t)} + H \gamma^{(t)} \gamma^{(8)}) & (t = 5, 6, 7), \\ \Omega_{x_4} &= \Omega_{x_8} = 0. \end{aligned}$$

It makes the gammas covariantly constant, $D_\mu \gamma^\nu = \partial_\mu \gamma^\nu + \Gamma^\nu{}_{\mu\lambda} \gamma^\lambda + [\Omega_\mu, \gamma^\nu] = 0$ for all 64 pairs (`gamma_covariantly_constant`, `covariant_constancy_D_mu_gamma_nu`), and its curvature is the Riemann tensor, $[D_\mu, D_\nu] \Psi = \frac{1}{2} R_{ab\mu\nu} S^{ab} \Psi$ (`spin_curvature_equals_Riemann`, `spinor_curvature_equals_riemann`).

The contraction that enters the field equation, per direction (no sum; formula key `gammaOmega_per_direction`):

$$\begin{aligned}
\gamma^{x_i} \Omega_{x_i} &= \frac{1}{2} a'_4 \gamma^{(4)} + \frac{1}{2} H \gamma^{(8)} & (i = 1, 2, 3), \\
\gamma^{x_t} \Omega_{x_t} &= -\frac{1}{2} a'_4 \gamma^{(4)} + \frac{1}{2} H \gamma^{(8)} & (t = 5, 6, 7), \\
\gamma^{x_4} \Omega_{x_4} &= \gamma^{x_8} \Omega_{x_8} = 0,
\end{aligned}$$

and summed:

$$\gamma^\mu \Omega_\mu = 3H \gamma^{(8)} = \frac{1}{2\sqrt{|g|}} \partial_\mu (\sqrt{|g|} \gamma^\mu)$$

(`gammaOmega_equals_3H_gamma_x8`, `gammaOmega_x4_terms_cancel`, `gammaOmega_divergence_form`). The sympy verifier confirms each of these statements independently (`gamma_mu_Omega_mu_equals_3H_gamma_x8`, `time_terms_cancel_hidden_term_survives` and `divergence_of_sqrtg_gamma`). The field-equations branch verifies the same value again, in a Clifford representation of its own and with T16 (`ownrep_gravity_term`, `authorT16_gravity_term`). For each direction separately $\{\gamma^\mu, \Omega_\mu\} = 0$ (no sum), because Ω_μ contains only generators $S^{(\mu)b}$ (`gamma_mu_anticommutates_with_Omega_mu_no_sum`, `anticommutator_gamma_Omega_vanishes`).

5 The Lagrangian

The Lagrangian density (SPEC section 3), with an explicit mass term linear in m and quadratic in the spinor:

$$\begin{aligned}
\mathcal{L} &= \sqrt{|g|} \left[\frac{1}{2} (\bar{\Psi} \gamma^\mu D_\mu \Psi - (D_\mu \bar{\Psi}) \gamma^\mu \Psi) - m \bar{\Psi} \Psi - U(\bar{\Psi} \Psi) \right], \\
\bar{\Psi} &= \Psi^\dagger C, \quad S = \bar{\Psi} \Psi, \quad U(S) = \frac{\lambda}{2} S^2 \quad (\text{general } U \text{ where stated}).
\end{aligned}$$

Properties, proved exactly for an explicit Grassmann algebra (Wolfram: generators θ_1, θ_2 and their conjugates, $\Psi_A = \sum_k \phi_{A,k}(x) \theta_k$; sympy: an own jet super-algebra; `grassmann_algebra_structure`, `superalgebra_axioms`):

- $\mathcal{L}$ is real: $\mathcal{L}^* = \mathcal{L}$, where conjugation reverses the order of Grassmann factors and m, λ are real (`L_real_G`, `grassmann_lagrangian_real`; the control `grassmann_controls_not_vacuuous` shows that the unsymmetrised form is not real off shell).
- It differs from the unsymmetrised form by a total divergence (`L_total_divergence_to_unsymmetrised_G`, `grassmann_total_divergence_relation`):

$$\mathcal{L} = \sqrt{|g|} [\bar{\Psi} \gamma^\mu D_\mu \Psi - mS - U(S)] - \frac{1}{2} \partial_\mu (\sqrt{|g|} \bar{\Psi} \gamma^\mu \Psi).$$

- In this metric the spin connection drops out of $\mathcal{L}$ itself, because $\bar{\Psi} \{\gamma^\mu, \Omega_\mu\} \Psi = 0$ term by term (`L_spin_connection_drops_out_G`):

$$\mathcal{L} = \cos z \left[\frac{1}{2} \sum_{a=1}^8 \frac{1}{f_a} (\bar{\Psi} \gamma^{(a)} \partial_a \Psi - \partial_a \bar{\Psi} \gamma^{(a)} \Psi) - mS - U(S) \right].$$

It reaches the field equations nevertheless: varying the symmetrised kinetic term produces $\partial_\mu(\sqrt{|g|}\gamma^\mu) = 2\sqrt{|g|}\gamma^\mu\Omega_\mu$ (section 7).

- S is an even element of the Grassmann algebra and U is a polynomial in S ; in the two-generator test algebra $S^3 = 0$, so $U = u_1 S + \frac{\lambda}{2} S^2$ is the most general potential there (`grassmann_algebra_a_structure`).

Negative control: the notebook's real Majorana-type Lagrangian $L_g = \sqrt{|g|} \Theta^T C \gamma^\mu D_\mu \Theta$ with 16 real anticommuting components is a total derivative, $L_g = \frac{1}{2} \partial_\mu (\sqrt{|g|} \Theta^T C \gamma^\mu \Theta)$: all its Euler-Lagrange expressions vanish identically and it has no field equations; moreover $\Theta^T C \Theta = 0$ (`Majorana_Lg_total_derivative_grassmann`, `negative_control_majorana_grassmann_total_derivative`). For real commuting components the same expression is not a total derivative (`Majorana_Lg_commuting_control`, `negative_control_majorana_commuting_contrast`). This is why the Dirac-type Lagrangian with $\bar{\Psi} = \Psi^\dagger C$ is used.

6 The exact covariant field equations (Euler-Lagrange)

Theorem (Euler-Lagrange equations). For the Lagrangian of section 5 with an arbitrary potential U , the variation with respect to $\Psi^\dagger$ (left derivatives) gives $\sqrt{|g|} C [\gamma^\mu D_\mu \Psi - (m + U'(S)) \Psi]$ and the variation with respect to Ψ (right derivatives) gives the adjoint expression, for every component and every Grassmann generator. The field equations are

$$\gamma^\mu D_\mu \Psi = (m + U'(S)) \Psi, \quad (D_\mu \bar{\Psi}) \gamma^\mu = -(m + U'(S)) \bar{\Psi},$$

with $U'(S) = \lambda S$ for the quadratic potential (`EL_Psibar_G`, `EL_Psi_G`, `grassmann_euler_lagrange_psibar_variation` with the control that $m \rightarrow -m$ fails, `grassmann_euler_lagrange_psi_variation`). The adjoint equation is the Dirac conjugate of the field equation, so the two form one system (`adjoint_equation_is_Dirac_conjugate_G`, `grassmann_adjoint_equation_is_conjugate`).

Written out in the primordial field (`Dirac_operator_explicit_G`; formula key `field_equation`):

$$\begin{aligned} & e^{-a_4} \sin^{-1/6} z (\gamma^{(1)} \partial_1 + \gamma^{(2)} \partial_2 + \gamma^{(3)} \partial_3) \Psi + \gamma^{(4)} \partial_4 \Psi \\ & + e^{a_4} \sin^{-1/6} z (\gamma^{(5)} \partial_5 + \gamma^{(6)} \partial_6 + \gamma^{(7)} \partial_7) \Psi \\ & + \tan z \gamma^{(8)} \partial_8 \Psi + 3H \gamma^{(8)} \Psi = (m + U'(S)) \Psi, \end{aligned}$$

and the adjoint equation (formula key `adjoint_equation`):

$$\begin{aligned} & e^{-a_4} \sin^{-1/6} z (\partial_1 \bar{\Psi} \gamma^{(1)} + \partial_2 \bar{\Psi} \gamma^{(2)} + \partial_3 \bar{\Psi} \gamma^{(3)}) + \partial_4 \bar{\Psi} \gamma^{(4)} \\ & + e^{a_4} \sin^{-1/6} z (\partial_5 \bar{\Psi} \gamma^{(5)} + \partial_6 \bar{\Psi} \gamma^{(6)} + \partial_7 \bar{\Psi} \gamma^{(7)}) \\ & + \tan z \partial_8 \bar{\Psi} \gamma^{(8)} + 3H \bar{\Psi} \gamma^{(8)} = -(m + U'(S)) \bar{\Psi}. \end{aligned}$$

Since $(\gamma^{(4)})^2 = -1$, the field equation is equivalent to the first-order evolution equation (`evolution_form_G`)

$$\partial_4 \Psi = -\gamma^{(4)} \left[(m + U'(S)) \Psi - \sum_{a \neq 4} \frac{1}{f_a} \gamma^{(a)} \partial_a \Psi - 3H \gamma^{(8)} \Psi \right],$$

so the slices $x_4 = \text{const}$ are non-characteristic ($g^{44} = -1$). In the chiral split $\Psi = (\psi_-, \psi_+)$ (rows 1 to 8 with $\Gamma = -1$, rows 9 to 16 with $\Gamma = +1$) every gamma is block off-diagonal, $\gamma^{(a)} = \begin{pmatrix} 0 & \bar{\tau}_a \\ \tau_a & 0 \end{pmatrix}$ with $\bar{\tau}_8 = \tau_8 = I_8$, and the field equation is the pair (`block_form`; formula key `field_equation_blocks`)

$$\sum_a \frac{1}{f_a} \bar{\tau}_a \partial_a \psi_+ + 3H \psi_+ = V \psi_-, \quad \sum_a \frac{1}{f_a} \tau_a \partial_a \psi_- + 3H \psi_- = V \psi_+, \quad V = m + U'(S).$$

The 16 component equations, $(\gamma^\mu D_\mu \Psi)_A = V \Psi_A$ with $\psi_A = \Psi_A$ the components in the basis of the fixture gammas, $\epsilon_s = e^{-a_4} \sin^{-1/6} z = 1/f_1$ and $\epsilon_t = e^{a_4} \sin^{-1/6} z = 1/f_5$ (formula key `field_equation_components`; `Dirac_operator_explicit_G`, `EL_Psibar_G`; the publication test of this document re-derives them from the formula file and compares them with the independent sympy expressions `dirac_equation_components` of the sympy report): the eight equations of the $\Gamma = -1$ rows,

$$\begin{aligned} \epsilon_s(-\partial_1 \psi_{16} + \partial_2 \psi_{15} - \partial_3 \psi_{14}) - \partial_4 \psi_{14} + \epsilon_t(\partial_5 \psi_{15} + \partial_6 \psi_{16} + \partial_7 \psi_9) + (\tan z \partial_8 + 3H) \psi_9 &= V \psi_1 \\ \epsilon_s(-\partial_1 \psi_{15} - \partial_2 \psi_{16} + \partial_3 \psi_{13}) + \partial_4 \psi_{13} + \epsilon_t(\partial_5 \psi_{16} - \partial_6 \psi_{15} + \partial_7 \psi_{10}) + (\tan z \partial_8 + 3H) \psi_{10} &= V \psi_2 \\ \epsilon_s(\partial_1 \psi_{14} - \partial_2 \psi_{13} - \partial_3 \psi_{16}) + \partial_4 \psi_{16} + \epsilon_t(-\partial_5 \psi_{13} + \partial_6 \psi_{14} + \partial_7 \psi_{11}) + (\tan z \partial_8 + 3H) \psi_{11} &= V \psi_3 \\ \epsilon_s(\partial_1 \psi_{13} + \partial_2 \psi_{14} + \partial_3 \psi_{15}) - \partial_4 \psi_{15} + \epsilon_t(-\partial_5 \psi_{14} - \partial_6 \psi_{13} + \partial_7 \psi_{12}) + (\tan z \partial_8 + 3H) \psi_{12} &= V \psi_4 \\ \epsilon_s(-\partial_1 \psi_{12} + \partial_2 \psi_{11} - \partial_3 \psi_{10}) + \partial_4 \psi_{10} + \epsilon_t(-\partial_5 \psi_{11} - \partial_6 \psi_{12} - \partial_7 \psi_{13}) + (\tan z \partial_8 + 3H) \psi_{13} &= V \psi_5 \\ \epsilon_s(-\partial_1 \psi_{11} - \partial_2 \psi_{12} + \partial_3 \psi_9) - \partial_4 \psi_9 + \epsilon_t(-\partial_5 \psi_{12} + \partial_6 \psi_{11} - \partial_7 \psi_{14}) + (\tan z \partial_8 + 3H) \psi_{14} &= V \psi_6 \\ \epsilon_s(\partial_1 \psi_{10} - \partial_2 \psi_9 - \partial_3 \psi_{12}) - \partial_4 \psi_{12} + \epsilon_t(\partial_5 \psi_9 - \partial_6 \psi_{10} - \partial_7 \psi_{15}) + (\tan z \partial_8 + 3H) \psi_{15} &= V \psi_7 \\ \epsilon_s(\partial_1 \psi_9 + \partial_2 \psi_{10} + \partial_3 \psi_{11}) + \partial_4 \psi_{11} + \epsilon_t(\partial_5 \psi_{10} + \partial_6 \psi_9 - \partial_7 \psi_{16}) + (\tan z \partial_8 + 3H) \psi_{16} &= V \psi_8 \end{aligned}$$

and the eight equations of the $\Gamma = +1$ rows,

$$\begin{aligned} \epsilon_s(\partial_1 \psi_8 - \partial_2 \psi_7 + \partial_3 \psi_6) + \partial_4 \psi_6 + \epsilon_t(-\partial_5 \psi_7 - \partial_6 \psi_8 - \partial_7 \psi_1) + (\tan z \partial_8 + 3H) \psi_1 &= V \psi_9 \\ \epsilon_s(\partial_1 \psi_7 + \partial_2 \psi_8 - \partial_3 \psi_5) - \partial_4 \psi_5 + \epsilon_t(-\partial_5 \psi_8 + \partial_6 \psi_7 - \partial_7 \psi_2) + (\tan z \partial_8 + 3H) \psi_2 &= V \psi_{10} \\ \epsilon_s(-\partial_1 \psi_6 + \partial_2 \psi_5 + \partial_3 \psi_8) - \partial_4 \psi_8 + \epsilon_t(\partial_5 \psi_5 - \partial_6 \psi_6 - \partial_7 \psi_3) + (\tan z \partial_8 + 3H) \psi_3 &= V \psi_{11} \\ \epsilon_s(-\partial_1 \psi_5 - \partial_2 \psi_6 - \partial_3 \psi_7) + \partial_4 \psi_7 + \epsilon_t(\partial_5 \psi_6 + \partial_6 \psi_5 - \partial_7 \psi_4) + (\tan z \partial_8 + 3H) \psi_4 &= V \psi_{12} \\ \epsilon_s(\partial_1 \psi_4 - \partial_2 \psi_3 + \partial_3 \psi_2) - \partial_4 \psi_2 + \epsilon_t(\partial_5 \psi_3 + \partial_6 \psi_4 + \partial_7 \psi_5) + (\tan z \partial_8 + 3H) \psi_5 &= V \psi_{13} \\ \epsilon_s(\partial_1 \psi_3 + \partial_2 \psi_4 - \partial_3 \psi_1) + \partial_4 \psi_1 + \epsilon_t(\partial_5 \psi_4 - \partial_6 \psi_3 + \partial_7 \psi_6) + (\tan z \partial_8 + 3H) \psi_6 &= V \psi_{14} \\ \epsilon_s(-\partial_1 \psi_2 + \partial_2 \psi_1 + \partial_3 \psi_4) + \partial_4 \psi_4 + \epsilon_t(-\partial_5 \psi_1 + \partial_6 \psi_2 + \partial_7 \psi_7) + (\tan z \partial_8 + 3H) \psi_7 &= V \psi_{15} \\ \epsilon_s(-\partial_1 \psi_1 - \partial_2 \psi_2 - \partial_3 \psi_3) - \partial_4 \psi_3 + \epsilon_t(-\partial_5 \psi_2 - \partial_6 \psi_1 + \partial_7 \psi_8) + (\tan z \partial_8 + 3H) \psi_8 &= V \psi_{16} \end{aligned}$$

For the quadratic potential $V = m + \lambda S$ with $S = \Psi^\dagger C \Psi$, an even (bilinear) element; the equations hold in the Grassmann algebra, and after quantisation as operator equations (section 11).

7 Non-triviality [1]

Theorem 1 (non-triviality [1]). Hypotheses: the metric of section 2 with $H > 0$ and $0 < z < \pi/2$; a_4 any twice differentiable function of x_4 (a constant a_4 included); real m and λ (or any real potential U); the canonical spin connection of the diagonal vielbein; Ψ any `dirac16complex` configuration (Grassmann-valued components) that is not identically zero. Then:

1. the Euler-Lagrange operator contains the gravitational term $\gamma^\mu D_\mu \Psi - \gamma^\mu \partial_\mu \Psi = \gamma^\mu \Omega_\mu \Psi = 3H\gamma^{(8)}\Psi$, and $3H\gamma^{(8)}\Psi \neq 0$;
2. per direction, the time-direction terms of the three inflating and the three deflating directions cancel ($\frac{3}{2}a'_4 - \frac{3}{2}a'_4 = 0$), while the hidden-direction terms of all six directions add ($6 \cdot \frac{H}{2} = 3H$);
3. in addition the vielbein enters every derivative term, $\gamma^\mu \partial_\mu = \epsilon_s \gamma^{(i)} \partial_i + \gamma^{(4)} \partial_4 + \epsilon_t \gamma^{(t)} \partial_t + \tan z \gamma^{(8)} \partial_8$, none of whose factors is identically 1; because of the cancellation in 2, a_4 enters the field equations through these factors $e^{\mp a_4}$;
4. all gravitational terms vanish together only in flat 4+4 space: $\Omega_\mu = 0$ for every μ if and only if $a'_4 = 0$ and $H = 0$, and $H = 0$ lies outside the family (a degenerate limit); no choice of local frame removes Ω_μ , because its curvature is the Riemann tensor and $R^{x_8}_{x_8} = -6H^2 \neq 0$: the metric is never flat for $H > 0$.

Proof. (a) The connection components of section 4 are each a'_4 or H times a factor $\pm e^{\pm a_4} \sin^{1/6} z$ that does not vanish on $0 < z < \pi/2$; since the S^{ab} are linearly independent, the a'_4 -part of Ω_μ vanishes if and only if a_4 is constant and its H -part if and only if $H = 0$ (`Omega_vanishes_iff_a4prime_and_H_vanish`, `nontriviality_Omega_zero_iff_flat`). (b) Contracting with $\gamma^\mu = \gamma^{(\mu)}/f_\mu$ and using $(\gamma^{(i)})^2 = 1$, $(\gamma^{(t)})^2 = -1$ gives the per-direction values of section 4: the $\gamma^{(4)}$ coefficients are $+\frac{1}{2}a'_4$ for each of the three inflating and $-\frac{1}{2}a'_4$ for each of the three deflating directions and cancel, the $\gamma^{(8)}$ coefficients are $\frac{1}{2}H$ for all six and add to $3H$; hence $\gamma^\mu \Omega_\mu = 3H\gamma^{(8)}$, a constant matrix independent of a_4 , x_4 and x_8 (`gammaOmega_x4_terms_cancel`, `gammaOmega_equals_3H_gamma_x8`, `time_terms_cancel_hidden_term_survives`, `gamma_mu_Omega_mu_equals_3H_gamma_x8`). The same value follows from the divergence form $\frac{1}{2\cos z} \partial_8 (\cos z \tan z) \gamma^{(8)} = \frac{6H \cos z}{2\cos z} \gamma^{(8)} = 3H\gamma^{(8)}$ (`gammaOmega_divergence_form`, `divergence_of_sqrtg_gamma`). (c) $(\gamma^{(8)})^2 = I_{16}$ and $\gamma^{(8)}$ is a signed permutation matrix, so each component of $\gamma^{(8)}\Psi$ is $\pm$ a component of Ψ ; therefore $3H\gamma^{(8)}\Psi = 0$ implies $\Psi = 0$ for $H > 0$ (`nontriviality_1_dirac16complex`). (d) The term is part of the Euler-Lagrange equation although the connection drops out of $\mathcal{L}$ in this metric: the variation of the symmetrised kinetic term produces $\partial_\mu(\sqrt{|g|}\gamma^\mu) = 2\sqrt{|g|}\gamma^\mu \Omega_\mu$, and the Euler-Lagrange expressions equal $\sqrt{|g|}C(\gamma^\mu D_\mu \Psi - V\Psi)$ with the full covariant derivative (`L_spin_connection_drops_out_G`, `EL_Psibar_G`). (e) The curvature of Ω_μ is $\frac{1}{2}R_{ab\mu\nu}S^{ab}$ (`spin_curvature_equals_Riemann`) and $R^{x_8}_{x_8} = -6H^2$ (`never_flat_for_H_positive`); if Ω_μ vanished in some frame, the Riemann tensor would vanish. $H = 0$ is a degenerate limit of the metric (`degenerate_at_H_0`). This completes the proof.

Remarks. In the formal flat limit ($a'_4 = 0$ and $H = 0$ with the factors frozen) all gravitational terms vanish and the equation becomes the flat 4+4 Dirac-type equation, as the test [1] requires ("unless

we are in flat 4+4 spacetime"). The surviving term is not cosmetic: the hidden-direction operator $\tan z \partial_8 + 3H$ is antisymmetric for the measure $\cos z dx_8$, $\cos z [p(\tan z \partial_8 + 3H)q + ((\tan z \partial_8 + 3H)p)q] = \partial_8(\sin z pq)$, and without $3H$ the mode operator of the quantised field is not Hermitian (`good_sector_hermiticity_curved`; formula key `hidden_direction_hermiticity`). Non-triviality [2], for the commuting field `dirac16complex00`, has the same algebraic content (`nontriviality_2_dirac16complex00`) and is the subject of the companion document `Revision/docs/DIRAC16COMPLEX00_FIELD_THEORY.md`.

8 Self-consistency

The field equations of section 6 are self-consistent in the following exact senses.

- One system. The adjoint equation is the Dirac conjugate of the field equation: with $E = \gamma^\mu D_\mu \Psi - (m + U')\Psi$ and $\bar{E} = (D_\mu \bar{\Psi})\gamma^\mu + (m + U')\bar{\Psi}$, $E^\dagger C = -\bar{E}$ exactly (`adjoint_equation_is_Dirac_conjugate_G`, `grassmann_adjoint_equation_is_conjugate`).
- Well-posed evolution. The evolution form of section 6 holds exactly (`evolution_form_G`), and the Heisenberg equation of the quantised field reproduces it (`Heisenberg_equation_reproduces_field_equation`, `hamiltonian_form_and_heisenberg_equation`).
- Integrability. The gammas are covariantly constant and the spinor curvature is the Riemann tensor (section 4). The square of the Dirac operator obeys the Lichnerowicz identity (`Lichnerowicz_identity_G`, `lichnerowicz_identity_on_fields`, `lichnerowicz_contraction`):

$$(\gamma^\mu D_\mu)^2 \Psi = g^{\mu\nu} (D_\mu D_\nu \Psi - \Gamma^\lambda_{\mu\nu} D_\lambda \Psi) - \frac{R}{4} \Psi, \quad R = 6((a'_4)^2 - 7H^2),$$

so the second-order consequence of the field equations is consistent with the curvature.

- Conserved current. $J^\mu = -i\bar{\Psi}\gamma^\mu\Psi$ is real, and off shell $\partial_\mu(\sqrt{|g|}J^\mu) = -i\sqrt{|g|}(\bar{E}\Psi + \bar{\Psi}E)$, so $\nabla_\mu J^\mu = 0$ on shell (`current_real_G`, `current_conservation_identity_G`, `grassmann_n_current_conservation`). The charge density is $J^{x_4} = \Psi^\dagger B \Psi$, so the conserved charge $Q = \int \cos z \Psi^\dagger B \Psi d^7x$ is an indefinite form (`charge_density_is_Krein_form_G`).
- Bianchi-type identity of the energy-momentum tensor. For the tensor $T^\mu{}_\nu$ of section 9 the invariance of the action under coordinate changes gives, off shell and for each λ , $\partial_\mu(\sqrt{|g|}T^\mu{}_\lambda) - \sqrt{|g|}T^\mu{}_\alpha \partial_\lambda e^a{}_\mu = \sum_\phi \text{EL}_\phi \partial_\lambda \phi$ (`Noether_identity_diffeomorphisms_G`), and the invariance under the 28 local frame rotations makes the antisymmetric part of T a combination of Euler-Lagrange expressions (`Noether_identity_local_Lorentz_G`). On shell, therefore, T is symmetric and $\nabla_\mu T^\mu{}_\nu = 0$ for general solutions (`conservation_on_shell_general`); the symplectic side verifies $\nabla_\mu T^\mu{}_\nu = 0$ for all eight ν after the on-shell substitution, with a negative control (`grassmann_emt_conservation_on_shell`, `grassmann_emt_conservation_on_negative_control`). The gravitational side satisfies the matching identities $\nabla_\mu E_{(k)}{}^\mu{}_\nu = 0$ (section 13).

- Exact solutions as test points. For $U = 0$ and every a_4 , $\Psi = \sin^\alpha z \left(\cosh(kx_4) + \frac{\sinh(kx_4)}{k} M \right) \chi$ with constant χ , $M = -m\gamma^{(4)} + 3H(2\alpha + 1)\gamma^{(4)}\gamma^{(8)}$, $k^2 = 9H^2(2\alpha + 1)^2 - m^2$, solves the field equation, and conservation of T and J holds exactly there (`solution_matrix_square`, `exact_solution_x4_x8_G`, `exact_solution_family_x4_x8`); the scope of this test is stated in the report: at these solutions the pressures and the momentum density vanish and $\rho = mS$ is constant, so the general proof is the pair of Noether identities. An exact nonlinear homogeneous solution with $U = \frac{\lambda}{2}S^2$ exists in the explicit Grassmann algebra, $\Psi = \sum_k [e^{M_m x_4} + \lambda S_0 \partial_m e^{M_m x_4}] \chi_k \theta_k$ with $M_m = -m\gamma^{(4)} + 3H\gamma^{(4)}\gamma^{(8)}$ (`exact_solution_nonlinearm_homogeneous_G`; formula key `exact_solutions`).

9 The energy-momentum tensor

The energy-momentum tensor is defined by the vielbein variation of the action and has the closed form (`T_vielbein_variation_closed_form_G`; formula key `T_variation`; sympy `grassmann_empt_equals_general_vielbein_variation_on_shell` with all 64 components of a general first-order vielbein variation)

$$T^\nu{}_\mu = e^b{}_\mu \frac{1}{\sqrt{|g|}} \frac{\delta S}{\delta e^b{}_\nu} = \delta^\nu{}_\mu \mathcal{L}_0 - \frac{1}{2} (\bar{\Psi} \gamma^\nu D_\mu \Psi - D_\mu \bar{\Psi} \gamma^\nu \Psi) - \frac{1}{4} g_{\mu\rho} \nabla_\lambda (\bar{\Psi} \{ \gamma^\lambda, \Sigma^{\nu\rho} \} \Psi), \quad \Sigma^{\nu\rho} = \frac{1}{4} [\gamma^\nu, \gamma^\rho], \quad \mathcal{L}_0 = \mathcal{L}/\sqrt{|g|},$$

where the last term (the totally antisymmetric spin density) comes from the variation of the spin connection. Its symmetric part is the Belinfante tensor (`T_symmetric_part_Belinfante_G`, `grassmann_empt_symmetric`; formula key `T_symmetric`),

$$T^{\nu\rho} = g^{\nu\rho} \mathcal{L}_0 - \frac{1}{4} (\bar{\Psi} \gamma^\nu D^\rho \Psi + \bar{\Psi} \gamma^\rho D^\nu \Psi - D^\rho \bar{\Psi} \gamma^\nu \Psi - D^\nu \bar{\Psi} \gamma^\rho \Psi),$$

which equals the variation tensor on shell (section 8). Sign convention (SPEC section 4): the energy density is $\rho = -T^{x_4}{}_{x_4}$ and the pressures are $p_\mu = T^\mu{}_\mu$ (no sum) for $\mu \neq x_4$; with this convention $\rho = mS + U$ for homogeneous on-shell configurations (section 10).

Kinetic and potential parts (formula key `EMT_kinetic_potential`): $T^\nu{}_\mu = T_{\text{kin}}{}^\nu{}_\mu + T_{\text{pot}}{}^\nu{}_\mu$ with

$$T_{\text{pot}}{}^\nu{}_\mu = -\delta^\nu{}_\mu (mS + U(S)),$$

$$T_{\text{kin}}{}^\nu{}_\mu = \delta^\nu{}_\mu \sum_\lambda K_\lambda - \frac{1}{4} (\bar{\Psi} \gamma^\nu D_\mu \Psi - D_\mu \bar{\Psi} \gamma^\nu \Psi + \bar{\Psi} \gamma_\mu D^\nu \Psi - D^\nu \bar{\Psi} \gamma_\mu \Psi).$$

Diagonal components in the primordial field (no sum; `T_diagonal_components_G`, `grassmann_empt_equals_vielbein_variation_diagonal`; formula key `EMT_diagonal`): with the directional kinetic terms

$$K_\mu = \frac{1}{2f_\mu} (\bar{\Psi} \gamma^{(\mu)} \partial_\mu \Psi - \partial_\mu \bar{\Psi} \gamma^{(\mu)} \Psi), \quad \mathcal{L}_0 = \sum_\mu K_\mu - mS - U(S),$$

$$T^\mu{}_\mu = \mathcal{L}_0 - K_\mu.$$

No spin-connection term survives in the diagonal components, because $\{\gamma^\mu, \Omega_\mu\} = 0$ for each μ . The off-diagonal component x_4 - x_8 , which SPEC section 5 lists among the components of the field equations for a_4 , is (T_x4_x8_component_G; formula key EMT_offdiagonal_x4_x8)

$$T^{x_4}_{x_8} = -\frac{1}{4}(B_{48} - \cot z B_{84}), \quad T^{x_8}_{x_4} = -\tan^2 z T^{x_4}_{x_8},$$

with $B_{48} = \bar{\Psi}\gamma^{(4)}\partial_8\Psi - \partial_8\bar{\Psi}\gamma^{(4)}\Psi$ and $B_{84} = \bar{\Psi}\gamma^{(8)}\partial_4\Psi - \partial_4\bar{\Psi}\gamma^{(8)}\Psi$; it vanishes for homogeneous on-shell configurations (grassmann_T_x4x8_homogeneous). The trace and the on-shell kinetic sum are (EMT_trace_G, grassmann_trace_on_shell, kinetic_sum_on_shell_G)

$$T^\mu{}_\mu = 7 \sum_\mu K_\mu - 8(mS + U), \quad \sum_\mu K_\mu = (m + U'(S))S + \frac{1}{2}(\bar{\Psi}E - \bar{E}\Psi),$$

so on shell $\mathcal{L}_0 = SU' - U$ and $T^\mu{}_\mu = -mS + 7SU' - 8U = -mS + 3\lambda S^2$. For a diagonal tensor $T = \text{diag}(p_3, p_3, p_3, -\rho, p_t, p_t, p_t, p_8)$ depending on x_4 only, conservation reads (energy_exchange_equation in both field-theory reports)

$$\nabla_\mu T^\mu{}_{x_4} = -\frac{d\rho}{dx_4} - 3a'_4(p_3 - p_t) = 0, \quad \nabla_\mu T^\mu{}_{x_8} = 3H \cot z (2p_8 - p_3 - p_t),$$

so energy flows between 3-space and the extra times unless $p_3 = p_t$ (the volume $\sqrt{|g|} = \cos z$ does not depend on x_4), and a conserved x_8 -independent diagonal tensor needs $p_8 = (p_3 + p_t)/2$.

For dirac16complex these expressions are even elements of the Grassmann algebra; they are the classical counterpart of the energy-momentum tensor operator of section 12, from which all numbers come.

10 Kinetic energy, potential energy, energy density, pressures and equations of state

From the diagonal components of section 9 (formula keys EMT_diagonal, EMT_kinetic_potential, equation_of_state_definitions):

$$\begin{aligned} \rho &= -T^{x_4}_{x_4} = \rho_{\text{kin}} + \rho_{\text{pot}}, & \rho_{\text{kin}} &= - \sum_{\mu \neq x_4} K_\mu, & \rho_{\text{pot}} &= mS + U(S), \\ p_3 &= T^{x_1}_{x_1} = \sum_{\mu \neq x_1} K_\mu - mS - U(S), \\ p_t &= T^{x_5}_{x_5} = \sum_{\mu \neq x_5} K_\mu - mS - U(S), \\ p_8 &= T^{x_8}_{x_8} = \sum_{\mu \neq x_8} K_\mu - mS - U(S), \end{aligned}$$

with $p_3 = T^{x_1}_{x_1} = T^{x_2}_{x_2} = T^{x_3}_{x_3}$ for configurations isotropic in 3-space and $p_t = T^{x_5}_{x_5} = T^{x_6}_{x_6} = T^{x_7}_{x_7}$ for configurations isotropic in the extra times. The kinetic energy density is ρ_{kin} , the potential

energy density $\rho_{\text{pot}} = mS + U$; the kinetic pressure in direction μ is $\sum_{\nu \neq \mu} K_\nu$ and the potential pressure is $-(mS + U)$ in every direction. The equations of state are

$$w_3 = \frac{p_3}{\rho}, \quad w_t = \frac{p_t}{\rho}, \quad w_8 = \frac{p_8}{\rho}.$$

Homogeneous on-shell configurations (depending on x_4 only): $K_\mu = 0$ for $\mu \neq x_4$ and $K_{x_4} = (m + U')S$, hence (`grassmann_homogeneous_on_shell_rho_p`, `exact_solution_nonlinear_homogeneous_G`; formula key `EMT_homogeneous_on_shell`)

$$\rho = mS + U, \quad p_3 = p_t = p_8 = SU' - U, \quad \rho_{\text{kin}} = 0, \quad p_{\text{kin}} = (m + U')S,$$

$$w_3 = w_t = w_8 = \frac{SU' - U}{mS + U},$$

and for $U = \frac{\lambda}{2}S^2$

$$\rho = mS + \frac{\lambda}{2}S^2, \quad p = \frac{\lambda}{2}S^2, \quad w_3 = w_t = w_8 = \frac{\lambda S}{2m + \lambda S}.$$

For the exact homogeneous solutions $S = S_0$ is constant along x_4 (`exact_solution_nonlinear_homogeneous_G`), so ρ , p and w are constant in x_4 , in agreement with the energy-exchange equation ($p_3 = p_t$ gives $d\rho/dx_4 = 0$). For `dirac16complex` all these quantities are Grassmann-algebra elements until they are evaluated as expectation values of the normal-ordered operators of section 12. The equation of state seen by a 3-space observer and its time dependence (the dark-sector hypotheses of SPEC section 8) are not computed in this document; they are the subject of `Revision/docs/DARK_SECTOR_HYPOTHESES.md`.

11 Canonical quantisation in 4 + 4 dimensions and the Krein structure

The evolution time is x_4 . The momentum conjugate to Ψ_A (right derivative) is $\pi_A = \frac{i}{2} \cos z (\Psi^\dagger B)_A$ for the symmetric Lagrangian and $i \cos z (\Psi^\dagger B)_A$ for the first-order form; it depends on $\Psi^\dagger$ only, a second-class constraint as for Dirac's field (`canonical_momentum` in both reports). Up to a total divergence $\mathcal{L} = \Psi^\dagger K \partial_4 \Psi -$ (terms without ∂_4) with $K = \cos z C \gamma^{(4)} = i \cos z B$, and the canonical (Dirac-bracket) anticommutator on a slice $x_4 = \text{const}$ is (`first_order_form_and_anticommutator`, `canonical_anticommutator_B`)

$$\{\Psi_A(x), \Psi_C^\dagger(y)\}_{x_4=y_4} = B_{AC} \frac{\delta^7(x-y)}{\sqrt{|g|}} = B_{AC} \frac{\delta^7(x-y)}{\cos z}.$$

With the Hamiltonian density $\mathcal{H} = \cos z [\Psi^\dagger (mC - \sum_{\mu \neq x_4} C \gamma^\mu D_\mu) \Psi + U(S)]$ (the unsymmetrised $\mathcal{L} = \Psi^\dagger K \partial_4 \Psi - \mathcal{H}$ exactly), the Heisenberg equation $\partial_4 \Psi = i[H, \Psi]$ equals the field equation solved for $\partial_4 \Psi$, with the operator ordering of the classical expression (`Heisenberg_equation_reproduces_field_equation`, `hamiltonian_form_and_heisenberg_equation`).

Theorem 2 (the canonical anticommutator forces a Krein structure). Hypotheses: the canonical anticommutator above, and $\Psi^\dagger$ realised as the adjoint of Ψ in a space with a positive-definite inner product. These are incompatible: there is no positive inner product with $\Psi^\dagger$ as the adjoint. *Proof.* B has the eigenvalue -1 with multiplicity 8; for the exact unit vector u with components $u_7 = -i/\sqrt{2}$, $u_{16} = 1/\sqrt{2}$ and all others zero, $Bu = -u$, and the smeared operator $X = \sum_A u_A^* \Psi_A(f)$ has $\{X, X^\dagger\} = (u^\dagger Bu) \|f\|^2 = -\|f\|^2 < 0$, whereas in a space with positive inner product $\{X, X^\dagger\} = XX^\dagger + X^\dagger X$ is a positive operator (`no_positive_inner_product` in both reports). Hence the state space carries an indefinite (Krein) inner product with fundamental symmetry B ($B = B^\dagger$, $B^2 = 1$, signature (8,8)); equivalently, in a positive Hilbert space the canonical conjugate $\Psi^\dagger$ is not the Hilbert adjoint χ but $\Psi^\dagger = \chi B$, with $\{\Psi_A, \chi_C\} = \delta_{AC}$.

The one-particle Krein structure. The Krein form $\int \cos z u^\dagger B v d^7 x$ is conserved by the evolution in the whole theory: the curved mode operator h ($i\partial_4 \Psi = h\Psi$ for $U = 0$) is self-adjoint for it (`Krein_form_conserved_curved`). For frozen coefficients (frame momenta k_a), plane waves $u e^{i(k \cdot x - E x_4)}$ obey $Eu = h_k u$ with $h_k = m\beta + \sum_a k_a \alpha^a$, $\beta = -i\gamma^{(4)}$, $\alpha^a = -\gamma^{(4)}\gamma^{(a)}$; β and $\alpha^{1,2,3,8}$ are Hermitian, $\alpha^{5,6,7}$ anti-Hermitian, $Bh_k = h_k^\dagger B$ (Krein self-adjoint), and (`mode_hamiltonian_good_sector`, `mode_hamiltonian_Krein_selfadjoint`, `mode_hamiltonian_B_selfadjoint_dispersion`)

$$h_k^2 = (m^2 + k_1^2 + k_2^2 + k_3^2 + k_8^2 - k_5^2 - k_6^2 - k_7^2) I_{16}.$$

Good sector (no dependence on the extra times, $k_5 = k_6 = k_7 = 0$): h_k is Hermitian with eigenvalues $\pm E$, eight each, and commutes with B , also with the curved term $3iH\gamma^{(4)}\gamma^{(8)}$; the curved mode operator is Hermitian for the measure $\cos z d^7 x$ exactly in this sector (`good_sector_spectrum_and_B_sectors`, `good_sector_hermiticity_curved`). Positive Fock realisation: $\Psi = \sum_s (b_s u_s + d_s^* v_s)$ with orthonormal eigenvectors u_s (energy $+E$) and v_s (energy $-E$), $\{b, b^*\} = \{d, d^*\} = 1$ on a positive Fock space, and $\Psi^\dagger = \chi B$, so that $\{\Psi_A, \Psi_C^\dagger\} = B_{AC}$. Exact examples (one momentum each): Wolfram, $m = 3$, $k = (4, 0, 0, 0)$ along x_1 , $E = 5$, on the explicit fermionic Fock space of 2^{16} occupation states; sympy, $m = 2$, $(k_1, k_2, k_3, k_8) = (1, 2, 0, 4)$, $E = 5$. The Hamiltonian $\chi h_k \Psi$ has the vacuum value $-8E$ (the filled sea; -40 in the sympy example), and after normal ordering the energy is $+E$ for the particle $b_0^*|0\rangle$ and $+E$ for the antiparticle $d_0^*|0\rangle$; the charge $\Psi^\dagger B \Psi = \chi \Psi$ is $+1$ and -1 (`Fock_space_good_sector_example`, `good_sector_positive_fock_realisation`).

Expectation-value rule. In the positive realisation, $\langle u | : \Psi^\dagger M \Psi : | u \rangle = u^\dagger B M u$ for a particle and $-v^\dagger B M v$ for an antiparticle (verified for $M = C$, $-iC\gamma^{(4)}$, $-iC\gamma^{(1)}$, $C\gamma^{(2)}\gamma^{(3)}$, a dense integer matrix and a generic symbolic matrix). In the Krein-Fock realisation ($\Psi^\dagger$ the Krein adjoint, modes with $u_n^\dagger B u_n = \epsilon_n$) the normalised expectation of $: \Psi^\dagger M \Psi :$ is $\epsilon_n u_n^\dagger M u_n$, which equals $u_n^\dagger B M u_n$ when $B u_n = \epsilon_n u_n$ and not for a Krein-boosted mode (`Fock_space_good_sector_example`, `good_sector_positive_fock_realisation`, `expectation_value_rule`).

Extra-time modes. With extra-time momentum, $E^2 = m^2 + k_s^2 - k_t^2$ is negative when $k_t^2 > m^2 + k_s^2$; exact example $m = 1$, $k_5 = 2$: $E = \pm i\sqrt{3}$, modes growing like $e^{\sqrt{3}x_4}$. In the metric the frame momentum $k_{(5)} = e^{a_4} \sin^{-1/6} z k_{x_5}$ grows as the extra times deflate (a_4 increasing), so every extra-time mode eventually enters the growing regime (a local-frame statement); extra-time momenta mix the $B = \pm 1$ sectors (`extra_time_modes_grow`, `mode_hamiltonian_good_sector`, `good_sector_spectrum_and_B_sectors`).

Scope: the positive Fock realisation is constructed and checked for single good-sector momenta with frozen coefficients; together with the exact Hermiticity of the curved good-sector mode operator and the conservation of the Krein form this is the quantisation recorded here. A complete construction

for the interacting field ($\lambda \neq 0$), for the extra-time sector with its growing modes, and of the field algebra on a whole slice is not part of this record (section 15).

12 The energy-momentum tensor operator

The energy-momentum tensor operator of `dirac16complex` is the tensor of section 9 with the field replaced by the quantised field and normal ordered with respect to the good-sector modes (formula key `quantisation`):

$$\hat{T}^\nu{}_\mu(x) = :T^\nu{}_\mu[\hat{\Psi}, \hat{\Psi}]:, \quad \hat{\bar{\Psi}} = \hat{\Psi}^\dagger C, \quad \hat{\Psi}^\dagger = \hat{\chi} B \quad (\text{positive realisation}).$$

Explicitly, with $\hat{S} = \hat{\Psi}^\dagger C \hat{\Psi}$ and $\hat{K}_\mu = \frac{1}{2f_\mu} (\hat{\bar{\Psi}} \gamma^{(\mu)} \partial_\mu \hat{\Psi} - \partial_\mu \hat{\bar{\Psi}} \gamma^{(\mu)} \hat{\Psi})$:

$$\begin{aligned} \hat{\rho} &= -\hat{T}^{x_4}{}_{x_4} = : \left(- \sum_{\mu \neq x_4} \hat{K}_\mu + m \hat{S} + U(\hat{S}) \right) :, \\ \hat{p}_\mu &= \hat{T}^\mu{}_\mu = : \left(\sum_{\nu \neq \mu} \hat{K}_\nu - m \hat{S} - U(\hat{S}) \right) : \quad (\mu \neq x_4), \\ \hat{T}^{\nu\rho} &= : \left(g^{\nu\rho} \hat{\mathcal{L}}_0 - \frac{1}{4} (\hat{\bar{\Psi}} \gamma^\nu D^\rho \hat{\Psi} + \hat{\bar{\Psi}} \gamma^\rho D^\nu \hat{\Psi} - D^\rho \hat{\bar{\Psi}} \gamma^\nu \hat{\Psi} - D^\nu \hat{\bar{\Psi}} \gamma^\rho \hat{\Psi}) \right) :, \end{aligned}$$

with the kinetic part $:(-\sum_{\mu \neq x_4} \hat{K}_\mu):$ and the potential part $:(m\hat{S} + U(\hat{S})):$ of the energy density. Properties and status:

- Normal ordering removes the vacuum value of the filled sea ($-8E$ per good-sector momentum in the exact examples of section 11) and gives positive energy $+E$ to particles and antiparticles (`Fock_space_good_sector_example`, `good_sector_positive_fock_realisation`).
- For $\lambda = 0$ every component of $\hat{T}$ is a normal-ordered bilinear $\hat{\Psi}^\dagger M \hat{\Psi}$ (with M containing the derivatives, which act on the mode functions), and its one-particle expectation values follow the expectation-value rule of section 11; in the sympy formulation the one-particle expectation of $\hat{T}$ is ϵ_n times the classical bilinear evaluated on the mode u_n (formula key `quantisation` of the sympy report).
- For $\lambda \neq 0$ the potential term is quartic. The field-equations record states the rule used there: every bilinear is replaced by the expectation value of its normal-ordered operator in the state, with $\langle :U(S): \rangle \neq U(\langle :S: \rangle)$ in general, and it records the operator form of the on-shell identity $\sum_\mu \langle :K_\mu: \rangle = \langle :(m + U')S: \rangle$ as an assumption to be confirmed by the theory branch (key `dirac16complex` of `Revision/field_equations_a4/a4-equations.json`). The identity is verified exactly for the Grassmann-algebra field (`kinetic_sum_on_shell_G`); its normal-ordered operator form for $\lambda \neq 0$ is not verified by a Revision check. Numerical expectation values of $\hat{T}$ for many-fermion states are computed only in the Kohn-Sham approximation (`Revision/kohn_sham/`, document `Revision/docs/KOHN_SHAM_DEFLATING_FIELD.md`) and are not quoted here.
- The Grassmann-algebra tensor of sections 9 and 10 obeys the identities of section 8 (symmetry and conservation on shell); the operator inherits them for the bilinear ($\lambda = 0$) part, where

they are linear in the field equations; for the quartic part this is part of the open operator question just stated.

13 The field equations for `a4[x4]` with `dirac16complex` as the source

Einstein-Lovelock gravity (SPEC section 5), with the Lovelock tensors computed with the generalised Kronecker delta in `Revision/gkd_lovelock` and recomputed in both verifiers of the field-equations branch (for example `P3_direct_equals_gkd_branch_monomials`, `P1_direct_equals_minus_4_Einstein` and `P4_vanishes_pigeonhole`; `sympy P3_equals_gkd_branch_monomials`):

$$\sum_{k=1}^3 \alpha_k E_{(k)}{}^\mu{}_\nu + \Lambda \delta^\mu_\nu = \kappa T^\mu{}_\nu, \quad E_{(k)} = -\frac{P_{(k)}}{2^{k+1}}, \quad E_{(1)} = G,$$

with $\alpha_1 = 1$ for Einstein gravity and the source $T^\mu{}_\nu = \text{diag}(p_3, p_3, p_3, -\rho, p_t, p_t, p_t, p_8) + q_{48} + q_{84}$ (q_{48} at $\mu = x_4, \nu = x_8$; q_{84} at $\mu = x_8, \nu = x_4$). The left-hand sides are divergence-free (`E1_divergence_free`, `E2_divergence_free`, `E3_divergence_free` in both reports). Their nonzero components are diagonal, with $x_1 = x_2 = x_3$ and $x_5 = x_6 = x_7$, free of x_8 and of a_4 itself; every off-diagonal left-hand side, including x_4 - x_8 , is zero (`independent_components`, `P1_structure`, `P2_structure`, `P3_structure`; `sympy other_components_vanish`). The diagonal components (formula key `lovelockTensors`; `json_lovelock_components`):

$$\begin{aligned} E_{(1)}{}^{x_1}{}_{x_1} &= -3(a'_4)^2 + a''_4 + 15H^2 \\ E_{(1)}{}^{x_4}{}_{x_4} &= 3(a'_4)^2 + 21H^2 \\ E_{(1)}{}^{x_5}{}_{x_5} &= -3(a'_4)^2 - a''_4 + 15H^2 \\ E_{(1)}{}^{x_8}{}_{x_8} &= -3(a'_4)^2 + 15H^2 \\ E_{(2)}{}^{x_1}{}_{x_1} &= 12(a'_4)^4 - 24(a'_4)^2 a''_4 + 168(a'_4)^2 H^2 - 40a''_4 H^2 - 180H^4 \\ E_{(2)}{}^{x_4}{}_{x_4} &= -36(a'_4)^4 - 120(a'_4)^2 H^2 - 420H^4 \\ E_{(2)}{}^{x_5}{}_{x_5} &= 12(a'_4)^4 + 24(a'_4)^2 a''_4 + 168(a'_4)^2 H^2 + 40a''_4 H^2 - 180H^4 \\ E_{(2)}{}^{x_8}{}_{x_8} &= 12(a'_4)^4 + 168(a'_4)^2 H^2 - 180H^4 \\ E_{(3)}{}^{x_1}{}_{x_1} &= -72(a'_4)^6 + 360(a'_4)^4 a''_4 - 648(a'_4)^4 H^2 + 432(a'_4)^2 a''_4 H^2 \\ &\quad - 1944(a'_4)^2 H^4 + 360a''_4 H^4 + 360H^6 \\ E_{(3)}{}^{x_4}{}_{x_4} &= 360(a'_4)^6 + 648(a'_4)^4 H^2 + 1080(a'_4)^2 H^4 + 2520H^6 \\ E_{(3)}{}^{x_5}{}_{x_5} &= -72(a'_4)^6 - 360(a'_4)^4 a''_4 - 648(a'_4)^4 H^2 - 432(a'_4)^2 a''_4 H^2 \\ &\quad - 1944(a'_4)^2 H^4 - 360a''_4 H^4 + 360H^6 \\ E_{(3)}{}^{x_8}{}_{x_8} &= -72(a'_4)^6 - 648(a'_4)^4 H^2 - 1944(a'_4)^2 H^4 + 360H^6 \end{aligned}$$

The independent field equations (formula key `generalSource`):

- constraint (x_4): $\sum_k \alpha_k E_{(k)}{}^{x_4}{}_{x_4} + \Lambda = -\kappa\rho$;
- 3-space ($x_1 = x_2 = x_3$): $\sum_k \alpha_k E_{(k)}{}^{x_1}{}_{x_1} + \Lambda = \kappa p_3$;

- extra times ($x_5 = x_6 = x_7$): $\sum_k \alpha_k E_{(k)}^{x_5}_{x_5} + \Lambda = \kappa p_t$;
- hidden direction (x_8): $\sum_k \alpha_k E_{(k)}^{x_8}_{x_8} + \Lambda = \kappa p_8$;
- off-diagonal: $0 = \kappa T^\mu_\nu$ for $\mu \neq \nu$, in particular $q_{48} = q_{84} = 0$.

The evolution equation is the 3-space minus extra-time combination; it contains a_4'' and factorises (`evolution_factorises_a4pp_times_F`, `evolution_factorises`; formula key `evolution_F`):

$$a_4'' F(a_4') = \kappa (p_3 - p_t),$$

$$F(a_4') = 2\alpha_1 + \alpha_2 (-48(a_4')^2 - 80H^2) + \alpha_3 (720(a_4')^4 + 864(a_4')^2 H^2 + 720H^4)$$

and F vanishes identically only for $\alpha_1 = \alpha_2 = \alpha_3 = 0$ (`evolution_F_not_identically_zero` in both reports). The constraint and the x_8 equation are first order (`x8_component_contains_no_a4pp`, `constraint_and_x8_first_order`). The remaining consistency conditions:

- algebraic condition: $\sum_k \alpha_k (E^{x_1}_{x_1} + E^{x_5}_{x_5} - 2E^{x_8}_{x_8}) = 0$ identically, so the source must satisfy $p_3 + p_t = 2p_8$ (`algebraic_identity_x1_plus_x5_minus_2x8`, `algebraic_identity`);
- x_8 -dependence: since the left-hand sides do not depend on x_8 , every component of T^μ_ν must be independent of x_8 , and $q_{48} = q_{84} = 0$ (key `x8_dependence`);
- constraint propagation (Bianchi identity): $\frac{d}{dx_4} \sum_k \alpha_k E_{(k)}^{x_4}_{x_4} = 3a_4' \sum_k \alpha_k (E^{x_1}_{x_1} - E^{x_5}_{x_5})$ identically (`constraint_propagation_bianchi`, `bianchi_x4`); with the conservation law $\rho' = -3a_4'(p_3 - p_t)$ the evolution equation follows from the constraint wherever $a_4' \neq 0$. The conservation components of a general source are $\nabla_\mu T^\mu_{x_4} = -\partial_4 \rho - 3a_4'(p_3 - p_t) + \partial_8 q_{84} - 6H \tan z q_{84}$ and $\nabla_\mu T^\mu_{x_8} = \partial_8 p_8 + 3H \cot z (2p_8 - p_3 - p_t) + \partial_4 q_{48}$ (`conservation_components` in both reports).

Einstein gravity ($\alpha_1 = 1$, $\alpha_2 = \alpha_3 = 0$; `einstein_components`; formula key `einstein`):

$$\begin{aligned} 3(a_4')^2 + 21H^2 + \Lambda &= -\kappa\rho \\ -3(a_4')^2 + a_4'' + 15H^2 + \Lambda &= \kappa p_3 \\ -3(a_4')^2 - a_4'' + 15H^2 + \Lambda &= \kappa p_t \\ -3(a_4')^2 + 15H^2 + \Lambda &= \kappa p_8 \end{aligned}$$

so $2a_4'' = \kappa(p_3 - p_t)$ and $\kappa(\rho + p_8) = -6((a_4')^2 + H^2) < 0$ for every a_4 and every Λ : the source must violate the null energy condition along the null direction $x_4 + x_8$ (`einstein_null_energy_x8`), and there is no vacuum solution for $H > 0$ (`einstein_no_vacuum_solution`, `einstein_no_vacuum`). Every twice differentiable a_4 is allowed if the source is the one these equations require (key `sourceRequiredByGivenA4`); $\kappa\rho > 0$ needs $\Lambda < -(3(a_4')^2 + 21H^2)$.

The linear member $a_4 = AHx_4 + a_0$ (3-space scale factor e^{AHx_4} , extra-time scale factor e^{-AHx_4} : exponential deflation for $A > 0$). The field equations then require $p_3 = p_t = p_8 = p$, with ρ and p constant, for any α_k (`linear_member_equal_pressures`; key `linearMember`):

$$\begin{aligned}
\kappa\rho &= \alpha_1(-3A^2H^2 - 21H^2) + \alpha_2(36A^4H^4 + 120A^2H^4 + 420H^4) \\
&\quad + \alpha_3(-360A^6H^6 - 648A^4H^6 - 1080A^2H^6 - 2520H^6) - \Lambda \\
\kappa p &= \alpha_1(-3A^2H^2 + 15H^2) + \alpha_2(12A^4H^4 + 168A^2H^4 - 180H^4) \\
&\quad + \alpha_3(-72A^6H^6 - 648A^4H^6 - 1944A^2H^6 + 360H^6) + \Lambda \\
\mathcal{V}(A) &= \alpha_1 + \alpha_2(-8A^2H^2 - 40H^2) + \alpha_3(72A^4H^4 + 144A^2H^4 + 360H^4)
\end{aligned}$$

A vacuum ($\rho = p = 0$) needs $\mathcal{V}(A) = 0$ (`linear_member_vacuum_factor`); in Einstein-Gauss-Bonnet gravity ($\alpha_3 = 0$) this gives $A^2 = (1 - 40\alpha_2H^2)/(8\alpha_2H^2)$, real for $0 < \alpha_2H^2 \leq 1/40$ (`einstein_gauss_bonnet_vacuum_linear`). For Einstein gravity $\kappa\rho = -(3A^2 + 21)H^2 - \Lambda$, $\kappa p = (15 - 3A^2)H^2 + \Lambda$ and $\kappa(\rho + p) = -6(A^2 + 1)H^2$.

Specialisation to `dirac16complex`. The source is the energy-momentum tensor of section 9 with the declared sign $\sigma_T = +1$ (key `emtConvention`: $T^\mu{}_\nu = \sigma_T(-K^{(\mu}{}_{\nu)} + \delta^\mu{}_\nu \mathcal{L}_0)$, the convention $\rho = -T^{x_4}{}_{x_4}$ of SPEC section 4; the relative sign of the gravitational and matter actions is carried as the parameter σ_T in the field-equations record and is not fixed by a Revision check). For the quantised field every bilinear is replaced by the expectation value of its normal-ordered operator in the state (section 12), with $k_\mu = K_\mu$:

$$\begin{aligned}
\rho &= \langle :k_4: \rangle - \langle :\mathcal{L}_0: \rangle, & p_3 &= \langle :\mathcal{L}_0: \rangle - \langle :k_1: \rangle, \\
p_t &= \langle :\mathcal{L}_0: \rangle - \langle :k_5: \rangle, & p_8 &= \langle :\mathcal{L}_0: \rangle - \langle :k_8: \rangle,
\end{aligned}$$

and the equations become (key `dirac16complex` of the field-equations record):

- constraint: $\sum_k \alpha_k E_{(k)}^{x_4}{}_{x_4} + \Lambda = -\kappa(\langle :k_4: \rangle - \langle :\mathcal{L}_0: \rangle)$;
- evolution: $a_4'' F(a_4') = \kappa(\langle :k_5: \rangle - \langle :k_1: \rangle)$: a state with 3-momenta and extra-time momenta has $\langle :k_1: \rangle \neq \langle :k_5: \rangle$ in general and then drives a_4'' ;
- hidden direction: $\sum_k \alpha_k E_{(k)}^{x_8}{}_{x_8} + \Lambda = \kappa(\langle :\mathcal{L}_0: \rangle - \langle :k_8: \rangle)$, and the algebraic condition $2\langle :k_8: \rangle = \langle :k_1: \rangle + \langle :k_5: \rangle$;
- every off-diagonal expectation value $\langle :T^\mu{}_\nu: \rangle$, $\mu \neq \nu$, must vanish, and every expectation value must be independent of x_8 . For a homogeneous configuration the off-diagonal components are multiples of the 15 three-gamma bilinears $\bar{\Psi}\gamma^{(a)}\gamma^{(b)}\gamma^{(c)}\Psi$ with $\{a, b, c\} = \{i, 4, 8\}$ ($i = 1, 2, 3, 5, 6, 7$) and $\{i, 4, j\}$ ($i = 1, 2, 3, j = 5, 6, 7$), derived for the classical homogeneous condensate (`condensate_offdiagonal_are_three_gamma_bilinears`, `authorT16_condensate_offdiagonal_three_gamma`; key `offDiagonalConditions`); for `dirac16complex` their expectation values must vanish.

Statement (states with equal 3-space and extra-time kinetic terms; key `homogeneousSingleMode`). Hypotheses: a state whose normal-ordered expectation values depend on x_4 only, satisfy the off-diagonal conditions and have $\langle :k_1: \rangle = \langle :k_5: \rangle$ for every x_4 (for example all quanta at zero 3-momentum and zero extra-time momentum); a_4 twice continuously differentiable; $(\alpha_1, \alpha_2, \alpha_3) \neq (0, 0, 0)$. Then $a_4'' F(a_4') = 0$; F is a nonzero polynomial in a_4' with finitely many roots (`evolution_F_not_identically_zero`) and a_4' is continuous, so $a_4'' = 0$ on every interval and $a_4 = AHx_4 + a_0$: the exponentially deflating linear member is the only a_4 such a source allows (the

same argument as the theorem for the classical condensate, key `theoremLinear`, with `linear_member_equal_pressures`). If in addition $\langle :k_\mu: \rangle = 0$ for every $\mu \neq x_4$, then $\rho = \langle :mS + U: \rangle$ exactly and $p = \langle :k_4: \rangle - \langle :mS + U: \rangle$, which equals $\langle :SU' - U: \rangle$ under the operator form of the on-shell identity of section 12 (an assumption for $\lambda \neq 0$). With it, Einstein gravity requires $\kappa \langle :mS + U: \rangle = -(3A^2 + 21)H^2 - \Lambda$ and $\kappa \langle :SU' - U: \rangle = (15 - 3A^2)H^2 + \Lambda$, hence $\kappa \langle : (m + U')S : \rangle = -6(A^2 + 1)H^2 < 0$ (key `einsteinConditions`; for the classical quadratic case `condensate_einstein_quadratic_U` in both reports).

What a_4 the equations allow, in summary: for a prescribed a_4 the equations fix the required source (ρ, p_3, p_t, p_8 above, with $p_3 + p_t = 2p_8$ and all components independent of x_8), and whether a state of `dirac16complex` supplies it is a separate question; states with $\langle :k_1: \rangle = \langle :k_5: \rangle$ allow only the linear member $a_4 = AHx_4 + a_0$ (exponential deflation of the extra times for $A > 0$) with constant ρ and p ; a non-linear a_4 needs $p_3 \neq p_t$, that is $\langle :k_1: \rangle \neq \langle :k_5: \rangle$, for example a Kohn-Sham gas whose quanta carry 3-momenta. The Kohn-Sham source ($\rho_{KS}, p_{3,KS}, p_{t,KS}, p_{8,KS}$ from the occupied orbitals, with the conditions above) is computed in `Revision/kohn_sham/` (key `kohnSham` of the field-equations record) and reported in `Revision/docs/KOHN_SHAM_DEFLATING_FIELD.md`, not here.

14 Relation to the pairing theorems

The pairing theorems of SPEC section 9 use the Lagrangian, the field equations and the energy-momentum tensor of this document; their full statements, hypotheses and limits are in `Revision/docs/PAIR_CREATION_PROOFS.md`. For `dirac16complex`, in brief:

- T1 (chirality): $\mathcal{L}_{m,\lambda}[\Gamma\Psi] = -\mathcal{L}_{-m,-\lambda}[\Psi]$, solutions of the (m, λ) equations are mapped to solutions of the $(-m, -\lambda)$ equations, $T \rightarrow -T$ and $J \rightarrow -J$, in every gravitational field; the total energy-momentum and the total current of the pair vanish identically (Wolfram `T1_Lagrangian_primordial_grassmann` and `T1_energy_momentum_primordial_grassmann`, sympy `T1.metric.grassmann.pair_total_emt_zero`).
- T2 (chirality combined with a $\text{Pin}(4,4)$ reflection of character -1 , the mirror $z \rightarrow \pi - z$ across the assumed Z2 brane): $(m, \lambda) \rightarrow (-m, \lambda)$, $\mathcal{L} \rightarrow +\mathcal{L}$, with equal, not opposite, energy-momentum (`T2_mirror_Lagrangian_grassmann`, `T2_mirror_energy_momentum_and_current_grassmann`; sympy `T2.metric.grassmann.emt`).
- Quantum reading: the chirality image $\Gamma\Psi$ carries the Krein metric $-B$; there is no cancellation between two independently quantised universes (`Q_Krein_metric_of_images`, `Q_no_identification_of_independent_universes`; sympy `Q.no_cancellation_independent_universes`).

These are correspondences between solutions; no creation process, rate or amplitude follows from them (`not_established` in the sympy pairing report).

15 What this document does not claim

- It derives no creation process, rate or amplitude for universes, and no statement that universes of masses $+m$ and $-m$ are created in pairs; the pairing theorems (section 14) map solutions to solutions.

- Non-triviality [1] is a statement about the Euler-Lagrange equations in the author's metric with $H > 0$; it is not a statement about observable effects of the gravitational term.
- The quantisation recorded here is: the canonical anticommutator, the theorem that it forces a Krein structure, the Krein self-adjointness of the evolution, and positive Fock realisations checked for single good-sector momenta with frozen coefficients. A positive-norm Hilbert space for the full field, a stable vacuum in the presence of the growing extra-time modes, and the interacting ($\lambda \neq 0$) operator theory are not constructed. In flat 4+4 space every energy eigenspace of the one-particle generator has Krein inertia (4,4) (Q_one_particle_Krein_signatures, Q.one_particle_Krein_inertia); the good-sector restriction removes the complex frequencies, not this indefiniteness.
- The normal-ordered operator form of the on-shell trace identity for $\lambda \neq 0$ is an assumption of the field-equations record, not a verified result; no numerical expectation value of the energy-momentum tensor operator is quoted in this document.
- The relative sign σ_T between the gravitational and the matter side of the a_4 equations is a declared convention ($\sigma_T = +1$), not derived.
- No solution $a_4(x_4)$ with a dirac16complex source is computed here beyond the statements of section 13; it is not claimed that any state of dirac16complex supplies a source that satisfies all the conditions of section 13 (in Einstein gravity such a source must violate the null energy condition along $x_4 + x_8$).
- The equation of state seen by a 3-space observer, its time dependence and the dark-sector hypotheses are not treated here.
- $H = 0$ is a degenerate limit, not a member of the family; the surface $z = \pi/2$ is degenerate ($g_{88} = 0$, $\sqrt{|g|} = 0$); nothing is claimed there.
- The comparison record of the field-theory branch is not used as evidence (section 1); every formula is cited by the separate checks of the two engines.

16 Reproduction

From the repository root (each verifier writes its outputs to the paths listed in section 1; the Wolfram verifier of each part runs first because the sympy verifier compares with its output at the end):

```
wolframscript -file Revision/algebra/wolfram/verify_algebra.wls
python Revision/algebra/python/check_algebra.py
wolframscript -file Revision/theory/wolfram/verify_field_theory.wls
python Revision/theory/python/check_field_theory.py
wolframscript -file Revision/field_equations_a4/wolfram/verify_field_equations_a4.wls
python Revision/field_equations_a4/python/check_field_equations_a4.py
wolframscript -file Revision/pairing/wolfram/verify_pairing.wls
python Revision/pairing/python/check_pairing.py
```

The sympy field-theory checker can write its report to another path, which leaves the recorded report untouched; it ran in about two minutes in this session:

```
python Revision/theory/python/check_field_theory.py --out <path>
```

This document (md, tex, pdf) is built and verified with the first command below (one command line); with the optional last argument it records the edition dirac16complex-field-theory in `Revision/pdf-specifications.json`. The second command runs the publication test, the third prints the blocks that the test generates:

```
python scripts/build_provenance_pdf.py Revision/docs/DIRAC16COMPLEX_FIELD_THEORY.md
--developer-layout --specifications Revision/pdf-specifications.json [--register]
python -m unittest Revision/tests/test_dirac16complex_field_theory_publication.py -v
python Revision/tests/test_dirac16complex_field_theory_publication.py --emit
```

The test checks that the .tex is the builder output of the .md, that the registered PDF is the PDF file next to it, that every cited check exists and passes in the reports, that the quoted check counts are those of the reports, and that the component field equations and the a_4 equations above are the output of its renderers applied to `Revision/theory/field-theory.json` and `Revision/field_equations_a4/a4-equations.json`.

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